

Optimization of daily scheduling for home health care services

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Abstract. As demographic trends show, the demand for home health care services will rise in future. Currently, the routing of the nurses is performed manually by the main service providers in Austria. This leads to a time-consuming process with a presumably suboptimal outcome. We present a model formulation and a metaheuristic solution approach, based on Variable Neighborhood Search, for optimizing the daily scheduling of the nurses. The objective function minimizes the traveling time of the nurses and the dissatisfaction level of clients and nurses. A feasible solution has to observe working time regulations, hard time windows, mandatory breaks, and a feasible assignment of nurses to clients. The proposed method finds the global optimal solutions for small problem instances. In extensive numerical studies it is shown that the algorithm is capable to solve real life instances with up to 512 home visits and 75 nurses. A comparison with a real life routing plan shows large savings potentials.

Keywords: vehicle routing; home health care; metaheuristics; variable neighborhood search

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Introduction

A lot of people require consistent medical treatment at their homes. Hence, the demand for home health care (HHC) services is growing. Reasons for this trend are manifold. Woodward *et al* (2004) present three possible explanations, they observed in Canada, that are also relevant for other industrial countries. First, more people suffer from chronic illnesses or physical disabilities. Second, people recovering surgery or acute illnesses often need further treatment at home. Third, the number of frail elderly people is rising due to the higher life expectancy. These people do need consistent health care service and support to remain at home. Schneider *et al* (2006) confirmed that in 2002 80,000 frail people needed HHC services in Austria, which range from qualified nursing to assistance in housekeeping.

We propose a model formulation and a solution approach, based on the metaheuristic Variable Neighborhood Search (VNS), for the daily planning of HHC services. The focus of the optimization lies in minimizing the traveling times of the nurses and the dissatisfaction level of clients and nurses. The traveling times thereby consist of the driving times and waiting times that arise if nurses are too early at the homes of the clients. The following constraints have to be observed to obtain a feasible solution: suitable assignments of nurses to clients (depending on qualification levels, language skills, and declinations), working time regulations, hard time windows, and mandatory breaks.

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In extensive numerical studies the solution approach has been verified to find good solutions. For small test instances, containing 20 clients and 4 nurses, the model is solved with solver software Xpress. The developed metaheuristic finds the global optimal solutions for these instances in short computation time. Moreover, the proposed algorithm solves real life instances with up to 512 jobs, 420 clients, and 75 nurses. The number of clients is smaller than the number of jobs, because some clients need several treatments per day. A comparison with a real life route plan shows large savings potential.

The interest in this research field is growing steadily. Begur *et al* (1997) and Cheng and Rich (1998) were probably the first who addressed this problem from an OR point of view. Begur *et al* (1997) developed a decision support system which is based on a savings heuristic and a nearest neighbor approach, whereas Cheng and Rich (1998) proposed a two and a three index formulation for HHC problems that minimize overtime and part-time work. They solved small instances with an exact approach and heuristically. Bertels and Fahle (2006) defined a hybrid of Constraint Programming, Simulated Annealing, and Tabu Search. Their objective consists of the minimization of the driving times and the maximization of the satisfaction of clients and nurses. Unlike the presented formulation, they did not consider breaks during the shifts. Eveborn *et al* (2006) developed a decision support system called Laps Care. Their problem is formulated as a set partitioning problem. Nurses are assigned to given working areas, so this problem can be subdivided into smaller subproblems. In their recent paper (Eveborn *et al*, 2009) they summarized the benefits of their approach on HHC in Sweden.

Bräysy *et al* (2007) computed the savings potentials for a case study in Finland and showed that it is possible to reduce the number of shifts up to 70%. Dohn *et al* (2008) propose a Branch and Price framework. However, the schedule of breaks is not included in their approach. The remainder of this paper is organized as follows: Section 2 gives a problem description and the mathematical model, whereas in Section 3 the solution approach is described. Section 4 presents empirical results on real life data and Section 5 concludes the paper.

Problem description

The demand for HHC is huge, but the number of nurses is not rising accordingly in Austria. Efficient route plans are therefore required to be able to maintain these services. Currently, the planning is done manually. This leads to an enormous effort and suboptimal tours. With the help of a decision support system the planning should improve in future. As a main service provider, the Austrian Red Cross has hereby an interest that the traveling times as well as the dissatisfaction level of clients and nurses are minimized. Together we defined six aims to measure the dissatisfaction of clients and nurses. As service is personal, clients often prefer to be visited by a particular nurse at a certain time. A lot of nurses work part-time and therefore would like to work at a certain time of the day. Overtime is usually also not preferred as well as visits to clients who require a lower qualification level. Additionally, some nurses do not get paid for the driving time to the first client and back home from the last one. Nevertheless, they have an interest in keeping these driving times small. These aims are combined in a weighted objective function where the weights sum up to 1. The choice of the weights rests with the decision maker(s).

Besides the complexity of the objective function, several constraints have to hold. For a proper description of these and the whole model the properties of the clients and nurses are given in the following.

Each client needs one or more treatments per day. These can include for example medication, insulin shots, and washing. Each treatment is declared as a job. Each job j has a certain qualification level q_j , duration d_j , and a time window $[a_j, b_j]$ in which the service has to start. Additionally, the preferred visiting time window is given by $[asoft_j, bsoft_j]$ and $a_j \leq asoft_j \leq bsoft_j \leq b_j$ holds. p_j is the preference of a client for a nurse. t_{ij} is the driving time between clients and starting (ending) points.

Each nurse n is defined by a qualification level Q_n and a contract working time S_n . After a working time of B a break is mandatory. The break duration $break_k$ as well as a desired time window for the break $[abreak_k, bbreak_k]$ can be specified by the nurse for each shift (tour) k he/she is working in. Generally, a nurse can work several shifts a day, as long as his/her total working time does not exceed the maximum allowed working time T and there is a break between the shifts at least as long as $break_k$. $v(k)$ equals the nurse of shift k . For each shift, the nurse would like to work from an_k to bn_k . This time window is soft. Due to different work contracts three possible starts of the tours are possible. The nurse might start at the depot D , or at home with payment of

the complete tour (H_1), or at home with payment from the arrival at the first client until the end of the visit of the last client (H_2).

Both clients and nurses have different language skills. Four languages with two options, speaking the language or not, are considered. Moreover, a nurse can reject servicing a specific client, and a client can decline a specific nurse, for any reason. A visit is therefore only allowed if the qualification level of the job is lower or equal the qualification level of the nurse, client and nurse speak the same language, and neither the client nor the nurse reject each other.

Mathematical model

Let x_{ijk} be a binary variable, indicating if job j is covered immediately after job i on tour k . y_k equals 1, if a nurse makes a break on tour k and s_{jk} equals the time at which the service is starting at job j on tour k . If a job is visited before (after) the beginning (end) of the soft time window, ltw_j (utw_j) equals the time between the service start and the beginning (end) of the soft time window for job j . $lntw_k$ and $untw_k$ measure the deviations of the nurses desired working time window for shift k . $lbtw_k$ and $ubtw_k$ are the time window violations of the desired break time and o_n equals the overtime of nurse n . J equals the set of jobs, N the set of nurses, V the set of shifts, and P is the break node. In a pre-processing step, it is checked whether a nurse $v(k)$ is allowed to treat a client's job j or not. If the answer is no, x_{ijk} is set to 0 in these cases before the optimization starts.

The following model has been developed:

$$\min \quad \alpha_1 \cdot \left(\sum_{k \in V} (en_k - sn_k - break_k \cdot y_k) - \sum_{j \in J} d_j \right) \quad (1)$$

$$+ \alpha_2 \cdot \sum_{n \in N} o_n \quad (2)$$

$$+ \alpha_3 \cdot \sum_{i \in J_0} \sum_{j \in J} \sum_{k \in V: p_j = v(k)} (1 - x_{ijk}) \cdot d_j \quad (3)$$

$$+ \alpha_4 \cdot \sum_{j \in J} (ltw_j + utw_j) \quad (4)$$

$$+ \alpha_5 \cdot \sum_{k \in V} (lntw_k + untw_k + lbtw_k + ubtw_k) \quad (5)$$

$$+ \alpha_6 \cdot \sum_{i \in J_0} \sum_{j \in J} \sum_{k \in V: q_j < Q_{v(k)}} d_j \cdot x_{ijk} \quad (6)$$

$$+ \alpha_7 \cdot \sum_{k \in H_2} \sum_{j \in J} (t_{v(k)j} \cdot x_{0jk} + t_{jv(k)} \cdot x_{j0k}) \quad (7)$$

Subject to

$$\sum_{i \in J_0} \sum_{k \in V} x_{ijk} = 1 \quad \forall j \in J \quad (8)$$

$$\sum_{j \in J} x_{jPk} = y_k \quad \forall k \in V \quad (9)$$

$$\sum_{j \in J} x_{0jk} \leq 1 \quad \forall k \in V \quad (10)$$

$$\sum_{j \in J} x_{j0k} \leq 1 \quad \forall k \in V \quad (11)$$

$$\sum_{i \in J_0} x_{ihk} = \sum_{j \in J_0} x_{hjk} \quad \forall h \in J, \forall k \in V \quad (12)$$

$$x_{Pjk} - x_{jPk} = 0 \quad \forall j \in J, \forall k \in V \quad (13)$$

$$x_{Pjk} - \sum_{i \in J} x_{ijk} \leq 0 \quad \forall j \in J, \forall k \in V \quad (14)$$

$$s_{ik} + t_{ij} + d_i - M_1 \cdot (1 - x_{ijk}) \leq s_{jk} \quad \forall i, j \in J, \forall k \in V \quad (15)$$

$$s_{Pk} + break_k - M_2 \cdot (1 - x_{jPk}) \leq s_{jk} \quad \forall j \in J, \forall k \in V \quad (16)$$

$$s_{ik} + t_{ij} + d_i - s_{Pk} \leq M_3 \cdot (2 - x_{ijk} - x_{jPk}) \quad \forall i, j \in J, \forall k \in V \quad (17)$$

$$sn_k + t_{0j} - M_4 \cdot (1 - x_{0jk}) \leq s_{jk} \quad \forall j \in J, \forall k \in D \quad (18)$$

$$sn_k + t_{v(k)j} - M_5 \cdot (1 - x_{0jk}) \leq s_{jk} \quad \forall j \in J, \forall k \in H_1 \quad (19)$$

$$sn_k - M_6 \cdot (1 - x_{0jk}) \leq s_{jk} \quad \forall j \in J, \forall k \in H_2 \quad (20)$$

$$s_{jk} + t_{j0} + d_j - M_7 \cdot (1 - x_{j0k}) \leq en_k \quad \forall j \in J, \forall k \in D \quad (21)$$

$$s_{jk} + t_{jv(k)} + d_j - M_8 \cdot (1 - x_{j0k}) \leq en_k \quad \forall j \in J, \forall k \in H_1 \quad (22)$$

$$s_{jk} + d_j - M_9 \cdot (1 - x_{j0k}) \leq en_k \quad \forall j \in J, \forall k \in H_2 \quad (23)$$

$$a_j \cdot \sum_{i \in J_0} x_{ijk} \leq s_{jk} \leq b_j \cdot \sum_{i \in J_0} x_{ijk} \quad \forall j \in J, \forall k \in V \quad (24)$$

$$s_{Pk} \leq \min(B + sn_k, M_{10} \cdot y_k) \quad \forall k \in V \quad (25)$$

$$\frac{en_k - sn_k - B}{B} \leq y_k \leq \frac{en_k - sn_k}{B} \quad \forall k \in V \quad (26)$$

$$en_k + break_k \leq sn_l \quad \forall k, l \in V: k < l, v(k) = v(l) \quad (27)$$

$$\sum_{k \in V: v(k)=n} (en_k - sn_k - break_k \cdot y_k) \leq T \quad \forall n \in N \quad (28)$$

$$o_n \geq \sum_{k \in V: v(k)=n} (en_k - sn_k - break_k \cdot y_k) - S_n \quad \forall n \in N \quad (29)$$

$$ltw_j \geq asoft_j - \sum_{k \in V} s_{jk} \quad \forall j \in J \quad (30)$$

$$utw_j \geq \sum_{k \in V} s_{jk} - bsoft_j \quad \forall j \in J \quad (31)$$

$$lbtw_k \geq (an_k + abreak_k) \cdot y_k - s_{Pk} \quad \forall k \in V \quad (32)$$

$$ubtw_k \geq s_{Pk} - (an_k + bbreak_k) \quad \forall k \in V \quad (33)$$

$$lntw_k \geq an_k - sn_k \quad \forall k \in V \quad (34)$$

$$untw_k \geq en_k - bn_k \quad \forall k \in V \quad (35)$$

$$o_n \geq 0 \quad \forall n \in N \quad (36)$$

$$ltw_j, utw_j \geq 0 \quad \forall j \in J \quad (37)$$

$$lbtw_k, ubtw_k, lntw_k, untw_k \geq 0 \quad \forall k \in V \quad (38)$$

$$x_{ijk} \in \{0,1\} \quad \forall i, j \in J_0 \cup P, \forall k \in V \quad (39)$$

$$y_k \in \{0,1\} \quad \forall k \in V \quad (40)$$

The objective function consists of (1) – (7). (1) gives the total traveling times of all nurses. (2) equals the overtime. The costs of unfulfilled preferences are added in (3). Soft time window violations of clients are represented in (4), whereas in (5) the soft time window violations of nurses (including preferred working times and breaks) are considered. In (6) service times of jobs covered by nurses with a higher qualification level than necessary are added. Unpaid driving times are considered in (7). Preferences and overqualified services are typically measured with the number of concerned jobs. The objective function however is given in minutes. Therefore, both preferences and overqualified services are multiplied by the service duration to obtain the same scale. This can also be interpreted as follows, the misspend time on clients who can/should be served by others is accounted.

(8) guarantee that each job is visited exactly once. If the break node P is visited in shift k , y_k equals 1, otherwise 0. Without loss of generality, node 0 represents the starting and ending point of all nurse's tours. However, the driving times from 0 to the first client (and back from the last client) depend on the true starting point of each nurse's tour. So, if a nurse is working and hence leaving node 0 (10), he/she also has to return to it (11). Additionally, if a nurse is visiting a client he/she also has to leave him/her again (12).

Breaks are made directly at the client's place. Therefore, the nurse "travels" to the break node and back to the client he/she came from (13). The time needed equals the break time. (14) assure that the break is made in the right tour. (15) – (17) guarantee that the starting times of jobs and breaks are set correctly. The start and end of each working time period are set accordingly in (18) – (23). $M_1 - M_{10}$ are hereby sufficiently large constants. Hard time window constraints of jobs are given in (24). (25) assure that the nurse starts her break after B minutes of working time at the latest. (26) guarantee that a break is made on a tour if and only if at least B minutes are

worked. If a nurse works several shifts a day, for instance in the morning and in the afternoon, the time between the shifts has to be at least the break time; otherwise, working time regulations would not hold (27). The total working time of a nurse (not including the break time) has to be smaller than T minutes (28).

In (29) – (35) relations of the decision variables are given. (29) state the overtime of the nurses, which is the surplus over the shift length S_n . (30) – (35) give the lower and upper soft time window violations of jobs, breaks, and nurses. Finally, (36) – (38) are non-negativity constraints and (39) – (40) are binary constraints.

The problem can be solved to optimality for 20 jobs, 20 clients, and 4 nurses with solver software Xpress. For real life instances it is not possible to find global optimal - or even feasible solutions with Xpress. Therefore, the metaheuristic approach presented in the next section is used.

Solution approach

A metaheuristic, based on Variable Neighborhood Search (VNS), was developed to solve real life scenarios with a size of up to 512 jobs, 420 clients, and 75 nurses. VNS was first mentioned by Mladenović and Hansen (1997), who also stated properties a metaheuristic should have in Hansen and Mladenović (2003). Among these are the following: simplicity, precision, coherence, efficiency, effectiveness, robustness, user-friendliness, and innovation. The proposed metaheuristic has most of these properties. Moreover, VNS is a widely used metaheuristic to solve vehicle routing problems. For example Polacek *et al* (2004), Hemmelmayr *et al* (2007), Fleszar *et al* (2009), and Imran *et al* (2009) present good and quick results for different VRP problems.

VNS is quite simple. Basically, given an initial solution, the neighborhood of it is explored for better solutions. If it is better, it is always accepted and the search starts over again with the new solution and the smallest neighborhood. If it is worse, it is rejected in the basic version and a larger neighborhood is explored. This procedure is repeated until some stopping condition, e.g. a certain number of iterations or a time limit is reached. For comparisons, the number of iterations is used as a stopping condition in the algorithm.

Variable Neighborhood Search (VNS) Algorithm

Initialization. Select the set of neighborhood structures $N_\kappa (\kappa = 1, \dots, \kappa_{\max})$ that will be used in the search; find an initial solution x ; choose a stopping condition;

Repeat the following until the stopping condition is met:

Set $\kappa \leftarrow 1$;

Repeat the following until $\kappa = \kappa_{\max}$:

Shaking. Generate a point x' at random from κ^{th} neighborhood of x ($x' \in N_\kappa(x)$)

Local search. Apply some local search method with x' as initial solution;

denote with x'' the so obtained local optimum;

Move or not. If this local optimum x'' is better than the incumbent, or if some acceptance criterion is met, move there ($x \leftarrow x''$) and continue the search with $N_1 (\kappa = 1)$ otherwise, set $\kappa \leftarrow \kappa + 1$;

Fig. 1. Steps of the VNS (c.f. Hansen and Mladenović (2001)).

If only better solutions are accepted during the search, it is likely to get trapped in local optima. Thus, often an acceptance criterion is used to decide whether a solution is accepted for further exploration or not. The scheme of the VNS algorithm is shown in Figure 1. In the following the steps of the developed algorithm are explained in detail.

Initial solution

A simple heuristic is used to obtain an initial solution for the algorithm. It works as follows: jobs are assigned to nurses who are allowed to serve them while time window restrictions as well as working time regulations are considered if possible. Therefore, all jobs are ordered according to the center of their time windows first. Afterwards, the driving times from the last position of each tour to the current job are ordered, beginning with the smallest. If the new working time is smaller than the shift length of the nurse and the time window holds, the job is added to

the tour and the driving times for the next job are considered. Otherwise, the next "closest" tour is tried out. If no tour fulfills both criteria, the job is added to the "closest" tour in which the job can be visited within its time window. If this is also not possible, the job is added to the "closest" tour with no overtime. If the job is still unassigned, it is added to the tour in which the working time has the smallest deviation from the shift length. The procedure stops when all jobs are assigned.

The starting times of the nurses are set in the way that the first client is visited at the beginning of his/her time window. If waiting times arise and all clients are visited within their time windows ($a_j \leq s_{jk} \leq b_j$), the tours are postponed to a later starting time to reduce waiting times. Breaks are set to some time between the visit of the first client (with waiting time) and the last client before the working time exceeds B minutes. The position is determined randomly. If two tours of a nurse overlap the second tour is postponed such that a break of at least $break_k$ occurs.

Shaking

This phase is the heart of the algorithm. The choice of the neighborhoods and the definition of how to explore them are crucial. Two different types of operators are used; the move and the cross-exchange operator. A more detailed explanation of them can be found in Van Breedam (1994), and Kindervater and Savelsbergh (1997).

Basically, the move operator moves a sequence of jobs from one tour to another tour. The sequence, as well as the tours are chosen randomly; but such that the segment of jobs can be treated by the new nurse. The sequence is uniformly distributed between 1 and the minimum of the current neighborhood size and the maximum tour length.

Cross-exchange is a combination of two moves in the sense that a segment is moved from tour 1 to tour 2 and a segment from tour 2 is moved to tour 1. Again, the assignment constraints have to hold. The tours and the segments are chosen randomly. The upper bound of the segment length depends on the maximum tour length and the current neighborhood size.

The size of the neighborhoods was determined with the help of a sensitivity analysis. Different combinations of move and cross exchange operators have been tested. A number of 4 move operators and a number of 8 cross exchange operators give the best results. Hence, in total 12 neighborhoods are used.

Local search

The obtained solution of the shaking step might not be optimal according the ordering of the jobs. The improvement heuristic 3-opt introduced by Lin (1965) is used to optimize the two changed tours. Basically, 3-opt exchanges 3 edges and replaces them by 3 new ones. A tour is usually accepted if it leads to a smaller total distance. In this approach we used a different concept of validation. The tour is first optimized according reduced waiting times and an optimal break position. Then the driving times and time window violations are compared to the previous tour. If both have improved the tour is accepted. The best improvement strategy was used.

Acceptance

In each iteration it has to be decided whether to keep the new solution or not. Plenty of possibilities exist. If only improving solutions are accepted during the search, one could easily get trapped in local minima. For diversification reasons it might be even beneficial to allow infeasible solutions temporarily. We defined an acceptance criterion consisting of two parts, a penalty function f and a probability measure.

If the new solution is better according to the penalty function (41) it is always accepted. Otherwise, a random number β between 0 and 1 is chosen. If it is smaller than the probability measure described in (42), the new solution is also accepted. The search starts then again with neighborhood $\kappa = 1$; if the solution is not accepted with neighborhood $\kappa + 1$. The penalty function consists of the objective function $z(x)$, the time window violations $g(x_k)$, and the working time violations $h(x_n)$.

$$f(x) = z(x) + \sum_{k \in V} \lambda_k \cdot g(x_k) + \sum_{n \in N} \mu_n \cdot h(x_n) \quad (41)$$

x is hereby the solution, x_k is the solution of tour k , and x_n is the solution of nurse n 's tour. λ_k and μ_n are dynamically adjusted weights. If the solution is feasible according to time windows (resp. working time) the weight is divided by δ . If it is infeasible, it is multiplied by this factor as long as the weights keep in the interval $[1, 10]$. The initial value for both is 10. The values of the boundaries of the interval as well as the initial value have been the result of various test runs.

$$\beta \leq \exp^{\log(\rho_1) + l \cdot \frac{\log(\rho_2) - \log(\rho_1)}{10^4}} \quad (42)$$

In (42) l equals the number of iterations, ρ_1 is the probability of accepting a solution straight at the beginning, whereas ρ_2 is the probability of accepting a solution after 10^4 iterations. The values of δ , the bounds of the interval, the starting value of 10, ρ_1 , and ρ_2 have been estimated with the help of a sensitivity analysis. δ is set to 1.001, ρ_1 equals 0.3, and ρ_2 is set to 0.1.

Empirical results

The algorithm was tested on randomly generated data sets as well as on real life data sets. The randomly generated data sets were used to benchmark the proposed method and to find good tuning parameters. A data set consisting of 100 jobs and 20 nurses was used to compute the parameters of our algorithm. The second data set consists of 20 instances with 20 jobs, 20 clients, and 4 nurses. The locations were spread randomly in a given area. The driving times are represented by a Euclidean distance matrix. Services have a qualification level between 1 and 3 and last for 30 minutes. The highest level is 3, for example an insulin shot. The lowest level is 1, which could be an aid in the household. Qualification levels, exclusions, language skills, and preferences are uniformly distributed. Time windows are chosen randomly over the whole day, with a width of 120 minutes for services with a qualification level of 3 and 240 minutes for the other two levels. The starting points of the nurses are also chosen randomly. S_n is uniformly distributed between 180 and 480 minutes. T equals 600 minutes, B 360 minutes, and the break time $break_k$ equals 30 minutes for all nurses. The soft time windows of the clients are set identically to the hard time windows, and the soft time windows of the nurses equal 720 minutes. The break time windows are set to $[240, 360]$. All nurses start their tours at the depot.

We solved the randomly generated data set with $\alpha_1 = 1$ with solver software Xpress 7.0 and the metaheuristic which was implemented in C++. The metaheuristic finds all global optimal solutions within 10^6 iterations. For 1 instance it was even found after 10^4 iterations and for 13 instances after 10^5 iterations. So, we conclude that the metaheuristic is capable of finding good results within short computation time. Bigger problem instances could not be solved with Xpress in reasonable computing time.

Real life tests

The Austrian Red Cross provided data of three regions in Upper Austria; one of them is urban (r_1) and two of them are rural (r_2, r_3). The first one is the smallest, containing 140 jobs, 140 clients, and 13 nurses. r_2 includes 351 jobs, 291 clients, and 39 nurses, whereas r_3 consists of 512 jobs, 420 clients, and 75 nurses.

ArcGIS was used to derive the driving times. Preferences and exclusions are non-existent and all nurses and clients are able to speak the same language (German). Information on the service times, time windows, and starting points is missing; hence they are estimated as follows. Service times are normally distributed, with a mean dependent on the qualification level of the job and a standard deviation of 15 minutes. It is reported that a job with a qualification level of 3 typically lasts 28.81 minutes, whereas a job with a qualification level of 1 needs 46.69 minutes on average. $S_n, T, B, break_k, a_j, b_j, abreak_k, bbreak_k$ are set as for the randomly generated data set. The soft time windows of the clients and the preferred working times of the nurses are chosen randomly. Considering

the starting points, four different scenarios are computed: All nurses are part of D , H_1 , H_2 , or a combination of them. For the latter the starting points were uniformly distributed.

The number of clients and jobs are given for a whole week. On average, clients need to be visited three times a week. The frequency of the jobs is normally distributed, with a mean of 3 and a standard deviation of 2. The jobs are then uniformly distributed among the five working days. Only in case of a frequency of 6 or 7 the weekends are considered.

Table 1, exemplary presents some results for the day with the most visits during the week for two parameter settings Λ_1 and Λ_2 . In the first (Λ_1), the objective equals the total traveling times with $\alpha_1 = 1$. In the second (Λ_2), $\alpha_1 = 0.5$, $\alpha_2 = 0.2$, $\alpha_3 = 0$, $\alpha_4 = 0.05$, $\alpha_5 = 0.05$, $\alpha_6 = 0.15$, and $\alpha_7 = 0.05$. These values are the results of a survey, in which decision makers from the Austrian Red Cross were asked to tell their preferences.

Table 1. Data and results (in minutes) for the day with the most visits during the week.

region	# jobs	# clients	# nurses	starting point	Λ_1	Λ_2
r_1	87	87	13	D	171	211
				H_1	201	255
				H_2	114	164
				$D \cup H_1 \cup H_2$	155	198
r_2	221	196	39	D	2,056	1,941
				H_1	1,982	1,889
				H_2	1,127	1,401
				$D \cup H_1 \cup H_2$	1,685	1,743
r_3	313	277	75	D	3,021	2,505
				H_1	2,944	2,759
				H_2	1,836	1,981
				$D \cup H_1 \cup H_2$	2,383	2,332

Overall, it can be observed that the best solution values are obtained when all nurses start their shifts at home without payment. This is not surprising, since the nurses do not get paid for the driving times spent to the first client and back from the last. It also shows that parameter setting Λ_1 is better in that special case. In the other cases it depends on the region and the choice of the starting points which parameter setup leads to a better solution value. For region r_1 all solution values obtained with Λ_1 are better than those obtained with Λ_2 . For region r_2 and r_3 this does not hold. Sometimes the solution values for Λ_1 are better, sometimes not. Typically, even if the solution value is better, this does not mean that each single aim is better fulfilled. Exemplary, Table 2 reports the detailed solution values for region r_2 . The traveling times are smaller for Λ_1 , but all other aims are better fulfilled for Λ_2 .

Table 2. Detailed results (in minutes) for the day with the most visits during the week for region r_2 .

Parts of objective function\scenarios	D		H_1		H_2		$D \cup H_1 \cup H_2$	
	Λ_1	Λ_2	Λ_1	Λ_2	Λ_1	Λ_2	Λ_1	Λ_2
solution value	2,056	1,941	1,982	1,889	1,127	1,401	1,685	1,743
traveling times	2,056	2,340	1,982	2,221	1,127	1,340	1,685	1,805
overtime	4,885	1,283	4,838	1,192	3,263	874	4,161	1,172
preferences	-	-	-	-	-	-	-	-
soft time window violations client	9,778	5,646	9,965	6,370	10,366	5,235	10,593	6,586
soft time window violations nurse	3,864	2,469	3,418	2,303	3,854	2,782	3,684	2,745
overqualification	3,653	724	3,160	710	2,979	528	3,184	761
unpaid driving times	-	-	-	-	1,525	1,519	569	506

Table 3 shows the results for an "extreme" day that means all clients in a district have to be visited on a single day. The results show that it is no problem in region r_1 to maintain services. For region r_2 and r_3 it is only possible to service all clients if the nurses start their shifts at home and do not get paid for their first and last trip. This means that under the assumption that all clients have to be visited on a single day within a time window of 120 to 240 minutes it is not possible to do so. However, such a scenario is quite unlikely. Nevertheless, it shows that under such circumstances more nurses are needed or the time windows have to be extended to uphold the services.

Table 3. Data and results for an "extreme" day.

region	# jobs	# clients	# nurses	starting point	Λ_1	Λ_2
r_1	140	140	13	D	338	428
				H_1	385	576
				H_2	248	423
				$D \cup H_1 \cup H_2$	301	457
r_2	351	291	39	D	∞	∞
				H_1	∞	∞
				H_2	∞	3,760
				$D \cup H_1 \cup H_2$	∞	∞
r_3	512	420	75	D	∞	∞
				H_1	∞	∞
				H_2	∞	5,356
				$D \cup H_1 \cup H_2$	∞	∞

Real life comparison

In 2010 the Austrian Red Cross gave us an actual route plan for region r_3 ; the data used in the earlier subsection is from 2008. The data set consists of 368 jobs, 293 clients, and 53 nurses. For one particular day the actual routing plan leads to a solution value of 3,703 minutes of traveling time. We got information about the service durations, visiting times, and starting points. The number of time-critical jobs is estimated to 10% of the jobs with a qualification level of 3 by the Austrian Red Cross. Jobs with a qualification level of 1 and 2 are usually not time-critical. In case that 10% of the jobs with a qualification level of 3 have to be visited within 120 minutes, a solution of 2,014 minutes can be obtained, which is an improvement of 45.6%. These solution values have been computed with parameter setup Λ_1 . Due to missing data the following results are all computed with this setup.

Table 4. Sensitivity analysis considering time-critical jobs and additional service times.

time-critical jobs\additional service time	0 %	10 %	20 %	30 %	40 %	50 %	60 %
0 %	1,855	1,920	2,087	2,197	2,310	2,625	2,708
10 %	2,014	2,031	2,235	2,309	2,519	2,770	3,257
20 %	2,051	2,147	2,241	2,467	2,641	2,903	∞
30 %	2,097	2,209	2,356	2,509	2,872	3,114	∞
40 %	2,157	2,330	2,427	2,800	2,999	∞	∞
50 %	2,249	2,342	2,726	2,804	3,259	∞	∞
60 %	2,312	2,579	2,787	3,144	∞	∞	∞
70 %	2,705	2,844	2,875	3,306	∞	∞	∞
80 %	2,726	2,928	2,983	3,450	∞	∞	∞
90 %	2,858	3,033	3,644	∞	∞	∞	∞
100 %	3,101	3,002	∞	∞	∞	∞	∞

Table 4 presents the results of a sensitivity analysis, in which the number of time-critical jobs with a qualification level of 3 and the service times of all jobs are increasing. This analysis was made because we were interested if it is possible to increase the service times and still get feasible solutions. The focus of our optimization lies mainly in reducing the traveling times which further leads to a reduction in working times. However, the working times of the nurses are constant. Therefore, more time can be spent at the serviced clients or new clients can be visited, which is the focus in the next analysis, presented in Table 5. The results in Table 4 show that all feasible solutions are better than the solution value of the current routing. However, if all jobs are time-critical it is not possible to find feasible solutions if the service time increases by 20%. Figure 2 shows these results as a graph. If the value is set to infinite (Table 4, Table 5, Figure 2, and Figure 3) no feasible solution was obtained.

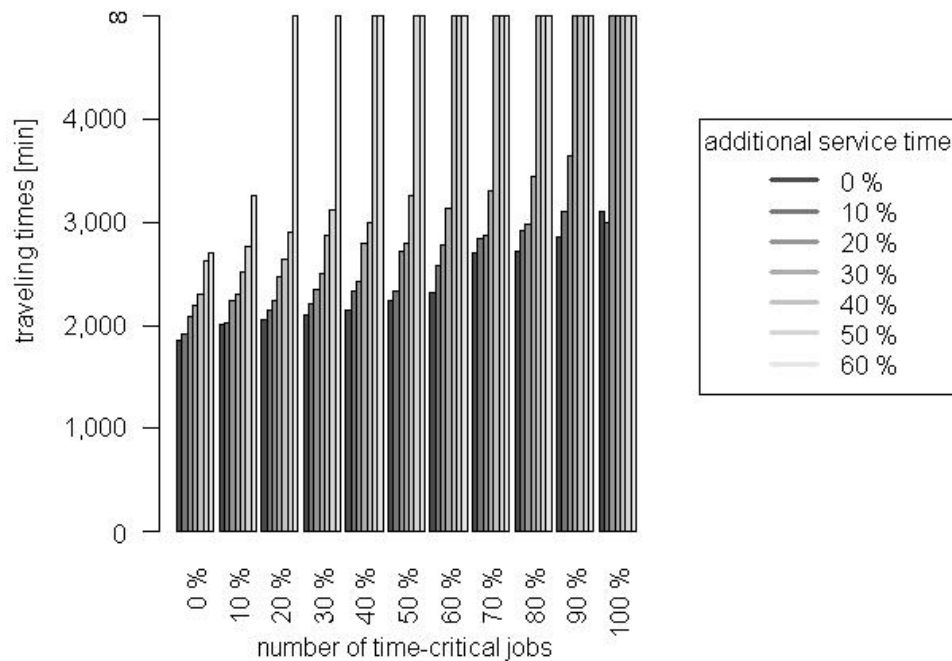


Fig. 2. Results of a sensitivity analysis with an increasing number of time-critical jobs and increasing service times.

Typically, some clients are on the waiting list for health care services. Hence, it is interesting, if it is possible to service at least some of them. Table 5 presents the results of a sensitivity analysis in which the number of time-critical jobs with a qualification level of 3 and the number of additional jobs are rising. The sample of additional jobs for each percentage of time-critical jobs is hereby independently drawn and chosen such that the percentage of each qualification level remains constant. In case of no time-critical jobs or 10% time-critical jobs a maximum number of 60% additional jobs can be included, whereas in the case of 100% time-critical jobs no jobs can be added. Figure 3 shows these results graphically.

Table 5. Sensitivity analysis considering time-critical jobs and additional jobs.

<i>time-critical jobs\additional jobs</i>	<i>0 %</i>	<i>10 %</i>	<i>20 %</i>	<i>30 %</i>	<i>40 %</i>	<i>50 %</i>	<i>60 %</i>
0 %	1,855	2,052	2,216	2,518	2,702	3,237	4,453
10 %	2,014	2,268	2,471	2,901	3,487	4,021	4,432
20 %	2,051	2,222	2,470	2,874	3,230	3,684	∞
30 %	2,097	2,353	2,735	3,084	3,308	∞	∞
40 %	2,157	2,579	2,881	3,409	3,851	∞	∞
50 %	2,249	2,621	2,934	3,826	∞	∞	∞
60 %	2,312	2,769	3,236	4,139	∞	∞	∞
70 %	2,705	3,017	3,339	4,560	∞	∞	∞
80 %	2,726	3,162	3,881	∞	∞	∞	∞
90 %	2,858	3,637	∞	∞	∞	∞	∞
100 %	3,101	∞	∞	∞	∞	∞	∞

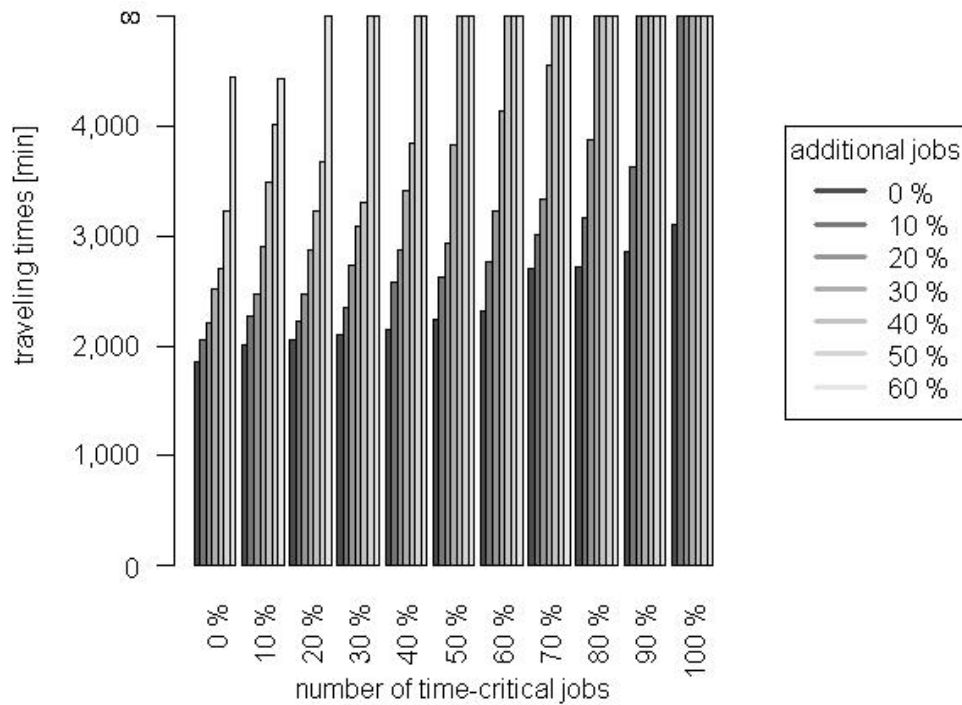


Fig. 3. Results of a sensitivity analysis with an increasing number of time-critical jobs and an increasing number of jobs.

Since we also assumed that more jobs are time-critical than in real life, it is possible to identify additional potential for the HHC providers. By using the proposed optimization method they could plan their visiting times more properly. Each client who requires a job with a qualification level of 3 could be assigned to a fixed time window. Therefore, he/she knows in advance for each visiting day the two-hour time window when the nurse appears. As the results show this procedure would not deteriorate the current solution.

Conclusion

This paper provides a new model formulation as well as a powerful VNS-based metaheuristic solution approach for optimizing the daily routing of HHC services. The proposed method was compared to the global optimal solutions obtained with solver software Xpress. It is proved that the VNS algorithm is capable to find the optimal solutions for small problem instances with 20 jobs (20 clients) and 4 nurses. Due to the high complexity, bigger problem instances are not solvable with Xpress in reasonable computing time. The proposed method was also applied to real life problem instances with up to 512 jobs (420 clients) and 75 nurses. It was tested for three areas and two different parameter settings in the objective function. The results show that in the urban area traveling times are much lower than in the two rural areas. In case that all visiting time windows are between 2 and 4 hours for all visits, it is possible to find feasible solutions for all starting point settings and regions for the day of the week with the most visits. If all clients have to be visited at one day, which is very unlikely to occur, it is possible to maintain the service in the urban area; due to the long driving times in the rural areas it is not possible with only one exception (setting Λ_2, H_2) to obtain feasible solutions. Nevertheless, a time window of 120 minutes for all jobs with a qualification level of 3 is not used in real life. It was introduced by us to show additional potential for the HHC providers in terms of client satisfaction. It is a huge advantage for the clients to know in advance fixed visiting time windows for each day. A comparison of the algorithmic results with an actual route plan shows that the current traveling time could be reduced by about 45%. Therefore, we decided to perform a sensitivity analysis where we increased the time-critical jobs, the service times, and the number of jobs. An increase in the number of time-critical jobs as well as an increase in service time leads to more satisfied clients. If the number of jobs increases, more people who are currently on the waiting list for HHC services could be served. Our analysis shows that these aims can be reached to a certain extent.

The paper shows that even a complex problem like the daily routing of HHC services, which has a lot of properties to keep in mind, can be solved efficiently. The current situation can be improved significantly and the numerical studies show some potential for the HHC service providers. Future research will concentrate on the important subject of solving the presented problem for a weekly time horizon. Another focus is put on the application of the presented method in natural disaster situations where quick solutions are necessary. Moreover, the Austrian Red Cross plans to use the proposed algorithm in its daily business.

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